



SECTION 1 — PYSPARK



DataFrame Fundamentals

- createDataFrame
- select/filter/withColumn
- alias/cast
- distinct/dropDuplicates
- union vs unionByName
- explode/split/regexp_extract
- when/otherwise



Joins

- inner/left/right/full/semi/anti joins
- broadcast joins
- skew join handling
- join optimization
- shuffle understanding
- Cartesian join risks



Window Functions

- row_number
- rank vs dense_rank
- lag/lead
- cumulative sum
- running totals
- moving averages
- partitionBy/orderBy
- rowsBetween/rangeBetween



Advanced Transformations

- recursive hierarchy handling
- flatten nested JSON
- pivot/unpivot
- dynamic columns
- metadata-driven transformations
- SCD Type 1 & 2



Performance Tuning

- repartition vs coalesce
- partition pruning
- predicate pushdown
- caching/persist
- AQE
- shuffle partitions
- spill handling
- skew handling
- explain plan reading



Spark Internals

- DAG
- stages/tasks/jobs
- narrow vs wide transformation
- driver vs executor
- Catalyst optimizer
- Tungsten
- lazy evaluation



SECTION 2 — SQL



Advanced SQL

- CTE
- recursive CTE
- window functions
- pivot/unpivot
- merge/upsert
- ranking problems
- gaps & islands
- consecutive streaks
- hierarchical queries



Query Optimization

- execution plan
- indexing concepts
- partitioning
- statistics
- join order

- anti-patterns

✅ Real Interview Problems

- duplicates removal
- nth highest salary
- cumulative salary
- call pair normalization
- latest record
- SCD queries
- recursive hierarchy

🔥 SECTION 3 — DATABRICKS

✅ Core Databricks

- clusters
- cluster policies
- job clusters vs all-purpose
- autoscaling
- pools
- notebook workflows
- widgets
- notebook.run
- dbutils

✅ Delta Lake

- ACID
- time travel
- vacuum
- optimize/z-order
- schema evolution
- merge
- CDC
- change data feed
- Delta transaction log

✅ Unity Catalog

- catalogs/schemas
- managed vs external tables

- external locations
- storage credentials
- lineage
- row filters
- masking
- RBAC
- audit logs

✅ Production Engineering

- Bronze/Silver/Gold
- schema drift handling
- metadata-driven framework
- reusable notebook design
- error handling
- retry strategy
- checkpointing
- idempotency

🔥 SECTION 4 — AZURE

✅ Azure Data Factory

- linked services
- datasets
- parameterization
- foreach
- dynamic content
- tumbling window
- triggers
- retry policies
- integration runtime

✅ Azure Storage

- ADLS Gen2
- hierarchical namespace
- storage tiers
- SAS vs Managed Identity
- lifecycle policies

✓ Azure Security

- Managed Identity
- Service Principal
- Key Vault
- RBAC
- firewall/private endpoints

✓ Monitoring

- Log Analytics
- Azure Monitor
- pipeline alerts
- Databricks logs
- SLA monitoring

🔥 SECTION 5 — REAL PRODUCTION SCENARIOS

✓ Must Be Able To Explain

- schema drift
- pipeline failure recovery
- late arriving data
- duplicate ingestion
- backfill strategy
- incremental load
- watermarking
- small file problem
- concurrent writes
- rollback strategy

✓ Critical Interview Scenarios

- column renamed suddenly
- datatype changed
- pipeline partially failed
- downstream dependency break
- performance degradation
- skew issue
- memory issue
- cluster sizing

SECTION 6 — STREAMING

Structured Streaming

- trigger modes
- watermark
- checkpoint
- exactly-once semantics
- deduplication
- stream-stream join
- stream-batch join

SECTION 7 — DEVOPS & CICD

DevOps

- Azure DevOps
- Git integration
- CI/CD pipelines
- deployment strategy
- environment promotion
- secrets management

SECTION 8 — ARCHITECTURE

Architecture Design

- medallion architecture
 - lakehouse
 - batch vs streaming
 - governance
 - cost optimization
 - scalability
 - disaster recovery
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 SECTION 9 — SOFT SKILLS

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 Senior-Level Expectations

- client communication
- requirement gathering
- mentoring juniors
- debugging production issues
- ownership mindset
- design discussions

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 SELF-ASSESSMENT SCORE

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Score Range	Level
0–80	Need preparation
80–140	Intermediate
140–200	Strong
200–260	Senior-ready
260+	Lead/Architect level